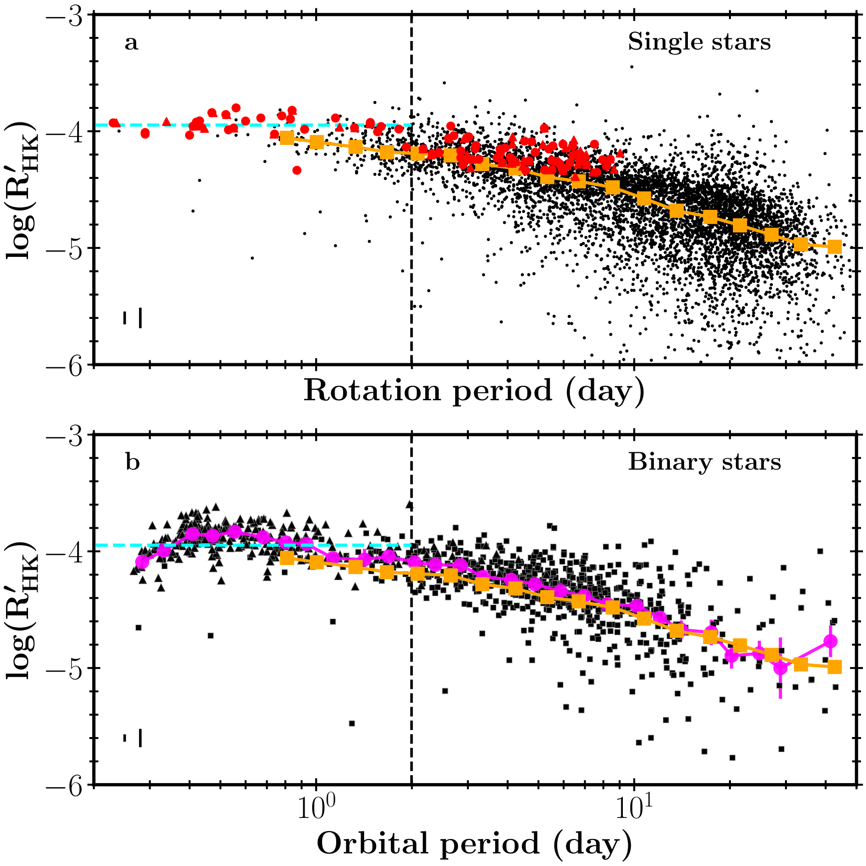

Supplementary Figure 1 | Comparison of magnetic activity between bright (*Gaia* G < 13) single and binary stars. Top: Period–activity diagram for *Kepler*/K2 single stars, with known binaries removed, and rotation periods adopted from the literature[1,2,3]. The red symbols represent a sample of reference stars observed by LAMOST and studied by Wright et al. (2011)[4] (red squares) and Fang et al. (2018)[5] (red triangles). The vertical dashed line at $P = 2$ days marks a conservative threshold below which magnetic activity is widely considered to be saturated[6]. The horizontal blue dashed line shows the mean $\log(R'_{HK})$ of the reference stars in the saturated regime, with the shaded region indicating twice the standard error. Orange squares connected by lines depict the mean $\log(R'_{HK})$ values within $\log P$ bins of width 0.11. The median formal $\log(R'_{HK})$ uncertainty and its conservative counterpart are shown in the lower-left corner (see Methods for details). **Bottom**: Similar to the top panel, except the horizontal axis shows orbital period for *Gaia* binary systems with circularised orbits (eccentricity < 0.03). Spectroscopic binaries are shown as squares, and eclipsing binaries as triangles. Magenta circles connected by lines represent mean $\log(R'_{HK})$ values within $\log P$ bins of width 0.11; their standard errors are generally too small to be visually discerned. The orange squares, connecting lines, and dashed line are identical to those in the top panel. The median formal and conservative $\log(R'_{HK})$ uncertainties for the binary sample are shown in the lower-left corner.

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